

Analysis PhD Exam, May 2010

Write solutions in a neat and logical fashion, giving complete reasons for all steps. Do any seven problems.

1. Consider Lebesgue measure on the real line $\mathbb{R}$. Give an example for each of the following, if possible. If not possible, give a brief explanation. A sequence f_n in $L^1(\mathbb{R})$ converging to an f in $L^1(\mathbb{R})$... Which of these answers change if we replace $\mathbb{R}$ by the interval $[0, 1]$?
 - a) ...in the L^1 norm but not in measure,
 - b) ...in measure but not in the L^1 norm,
 - c) ...in the L^1 norm but not a.e.,
 - d) ...a.e. but not in measure.
2. Define the Hardy–Littlewood maximal function. State and prove the Hardy–Littlewood maximal theorem. (Begin by proving an appropriate covering lemma.)
3. This problem has two parts.

- a) State Egorov's theorem.
- b) Suppose $(X, \mathcal{M}, \mu)$ is a measure space with $\mu(X) < \infty$ and $f : X \rightarrow \mathbb{C}$ is measurable. Let (f_n) be a sequence of integrable functions such that $f_n \rightarrow f$ a.e. Suppose further that the sequence f_n is uniformly integrable, that is, for every $\epsilon > 0$ there exists a $\delta > 0$ such that if E is measurable and $\mu(E) < \delta$, then

$$\int_E |f_n| d\mu < \epsilon.$$

Prove that f is integrable and

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0.$$

4. Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces. Suppose that (T_n) is a sequence of bounded linear operators from $\mathcal{X}$ to $\mathcal{Y}$ such that $\lim_{n \rightarrow \infty} T_n x$ exists for every $x \in \mathcal{X}$. Prove that

$$Tx := \lim_{n \rightarrow \infty} T_n x$$

defines a bounded linear operator from $\mathcal{X}$ to $\mathcal{Y}$.

5. This problem has two parts.
 - a) State the Lebesgue–Radon–Nikodym theorem.
 - b) State and prove a version of the chain rule for Radon–Nikodym derivatives.
6. Let X be a normed vector space. Given $x \in X$, define a linear functional $\hat{x} : X^* \rightarrow \mathbb{C}$ by $\hat{x}(f) = f(x)$. Prove that the map $x \rightarrow \hat{x}$ is an isometry from X into X^{**} .
7. Let $\mathcal{H}$ be a Hilbert space and suppose $\mathcal{M}, \mathcal{N}$ are closed subspaces with $\mathcal{M} \perp \mathcal{N}$. Prove that $\mathcal{M} + \mathcal{N}$ is closed.

8. Suppose $(k_n)_{n=1}^\infty$ is a sequence of functions in $L^1(\mathbb{R})$ satisfying the following. Prove that

$$\lim_{n \rightarrow \infty} \|f - f * k_n\|_1 = 0$$

for all $f \in L^1(\mathbb{R})$. (Here $*$ denotes convolution: $(f * g)(x) := \int_{-\infty}^{\infty} f(x-t)g(t) dt$.)

- a) $\int_{-\infty}^{\infty} k_n(t) dt = 1$ for all n ,
- b) $\sup_n \int_{-\infty}^{\infty} |k_n(t)| dt < \infty$, and
- c) for each $\delta > 0$, $\sup_{|t| \geq \delta} \{|k_n(t)|\} \rightarrow 0$ as $n \rightarrow \infty$.